

Is Family-Sized Housing Owner-Only? Evidence On The Tenure-Segmentation Assumption

Date: 2026-09-18. Status: evidence note. No model code was run and no model or calibration change is proposed here.

Lead review (September 18, 13:30). Two claims in this note are not verified by the lead and should be treated as the memo's, not the paper's, until checked. (1) The Kaplan, Mitman and Violante Table 6 renter shares by size class and the Section 5.3.1 “virtually indistinguishable” segmentation robustness are attributed to the published 2020 version; the NBER working paper (w23694) that I opened has the size grids $\{1.50, \dots, 5.15\}$ owner and $\{1.17, 1.50, 1.92\}$ rental, as stated, but neither the table nor the robustness passage. (2) The statement that the cap is “the model’s strongest fertility lever” (six to eight rooms: TFR +0.071, family-ownership gap 0.296 to 0.035) is taken from `output/model/fable_size_mapping_audit_20260701/HANDOFF_RESTART.md`, a July 1 audit on the earlier package at a coarse wealth grid whose mechanism numbers are recorded as unreliable. It has not been measured on the paper’s model. The direct test was run on September 19: with the cap at eight rooms and nothing else changed, completed fertility is 1.876 against 1.872 at fixed child preference, childlessness and first-birth age are unchanged, and ownership falls from 0.46 to 0.31. On the paper’s model the cap is an ownership lever, not a fertility lever, so the “strongest fertility lever” line below is withdrawn. The DUE finding (rental maximum equal to the owner maximum, 12.7, DUE.txt line 1211) is verified against the local text and does contradict the July cap note’s attribution.

(a) The question

The model gives renters and owners different size menus. A renter chooses housing services h continuously on the interval $(0, \bar{h}^R]$ with $\bar{h}^R = 6$ rooms; an owner chooses from the discrete ladder $\mathcal{H} = \{2, 4, 6, 8, 9.5, 11\}$ rooms. Both objects are set in `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26/code/model/intergen_housing_fertility/parameters.py` (lines 122-125). The consequence is that the three largest rungs, 8, 9.5 and 11 rooms, are reachable only by buying. Because the model’s child-space requirement $\bar{h}(n)$ rises with the number of children, a household that wants a third or fourth child must cross the down-payment threshold $(1 - \phi)$ to get the space. That is the paper’s central mechanism, so the question is not cosmetic: is “large units are owner-only” a constraint on the household’s choice set, or is it a compact description of an allocation that households would produce anyway through preferences, life-cycle wealth and landlord economics? A cap that merely describes the observed allocation is not a mechanism; a cap that binds is. The evidence below separates what is measured from what is assumed.

(b) What the paper’s own figure shows

The slide is the frame “Big Units Are Owned — Renting Is Not a Substitute” in `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26/latex/wedge2_hacamo_slides.tex` (lines 78-100). Both panels come from the 2023 American Housing Survey national public-use household file, version 1.1, weighted by WEIGHT, built by `/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26/code/data/ahs_supply_snapshot/analyze_ahs_family_unit_menu.R`. The weighted read is 145.3 million housing units, 133.2 million occupied. Size is the AHS bedroom count, binned 0-1, 2, 3, 4+. The left panel normalizes within tenure; the right panel normalizes within size bin, with owner, renter and vacant summing to 100 in each bin.

Object (AHS 2023, national)	Value
Share of owner-occupied units with 3+ bedrooms	81.3%
Share of renter-occupied units with 3+ bedrooms	30.2%
Owner share of 0-1 bedroom units	12.8%
Owner share of 2 bedroom units	39.4%
Owner share of 3 bedroom units	75.4%
Owner share of 4+ bedroom units	87.0%
Renter share of 4+ bedroom units	9.2%

Source file: `code/data/ahs_supply_snapshot/output_ahs_family_unit_menu_national/ahs_stock_by_bedroom_tenure.csv`. The companion frame reports that 79% of households with children live in 3+ bedroom units against 59% of childless households, and that the parent share within size bins rises 6, 21, 29, 41% across the four bins. I recomputed all of these from the underlying CSVs and they reproduce exactly.

Two things the figure does not show. It is a cross-section of occupancy, not of availability, so it cannot distinguish a choice-set restriction from sorting. And it is stated in bedrooms, while the model is stated in rooms; the room version of the same table is weaker, because 7+ room units are 89.0% owner, 8.0% renter and 3.0% vacant, and AHS does not publish a finer top bin.

(c) What the data say

Fact	Value	Source, year
Renter share, studio	86.5%	ACS 2024 1-year, B25042
Renter share, 1 bedroom	84.9%	ACS 2024 1-year, B25042
Renter share, 2 bedrooms	54.9%	ACS 2024 1-year, B25042
Renter share, 3 bedrooms	20.1%	ACS 2024 1-year, B25042
Renter share, 4 bedrooms	11.0%	ACS 2024 1-year, B25042
Renter share, 5+ bedrooms	8.3%	ACS 2024 1-year, B25042
Renters living in 3+ bedroom units	29.7%	ACS 2024 1-year, B25042
Owners living in 3+ bedroom units	80.5%	ACS 2024 1-year, B25042
Renter share, 1-unit detached structures	13.7%	ACS 2024 1-year, B25032
Renter share, 5+ unit structures	86.1%	ACS 2024 1-year, B25032
3+ bedroom share of units in 20+ unit buildings	7.6%	AHS 2023 national
3+ bedroom share of detached single-family units	83.5%	AHS 2023 national
Renter-occupied 3+ bedroom units, count	13.4 million	AHS 2023 national
Of those, share in single-family (detached + attached)	68.7%	AHS 2023 national

Fact	Value	Source, year
Single-family rental stock	14.9 million units, 2022; peak 16.1 million in 2016	JCHS, <i>America's Rental Housing 2024</i>
Single-family built-as-rental completions	26,000/yr in 2010 to 67,000/yr in 2022; 13% of new rental construction	JCHS 2024
Renter share, 3 bedrooms, New York metro	29.1%	ACS 2024 1-year, B25042
Renter share, 3 bedrooms, Los Angeles metro	28.2%	ACS 2024 1-year, B25042
Renter share, 5+ bedrooms, New York metro	8.9%	ACS 2024 1-year, B25042
Prime-age childless renters with 6+ rooms	13.8%	ACS/IPUMS 2012-2023, MMS metros
Prime-age childless renters with 7+ rooms	6.0%	ACS/IPUMS 2012-2023, MMS metros
Owner minus renter mean rooms, prime-age childless	2.42 rooms	ACS/IPUMS 2012-2023, MMS metros
Median gross rent by bedrooms (studio to 5+)	\$1,321 / \$1,301 / \$1,490 / \$1,677 / \$2,069 / \$2,182	ACS 2024 1-year, B25031
Implied rent per bedroom (1BR to 5+BR)	\$1,301 / \$745 / \$559 / \$517 / \$436	own arithmetic on B25031
AHS mean rent per square foot by bedrooms	\$2.09 / \$1.49 / \$1.18 / \$1.14	AHS 2023 national
Within-cell marginal rent premium above 5.5 rooms	\$7.7 per room (se \$15.6), statistically zero	ACS/IPUMS, 1,284,904 renter heads
Central 3+ bedroom rental price premium, metro-weighted	-0.02 log points	ACS/IPUMS 2012-2023

The ACS room and bedroom moments are in `code/data/mms_center_periphery/output_intergen_one_market_targets/intergen_one_market_acs_housing_targets.csv`. The within-cell rent regression is documented in `output/model/fable_size_mapping_audit_20260701/SOFT_CAP_DESIGN_MEMO.md`, section 4: rent on rooms with a kink, year by metro by location fixed effects, household weights, standard errors clustered by metro. Below the knot the marginal room is worth \$85.1 (se \$11.9); above it the extra premium is \$7.7 (se \$15.6) at a 5.5 room knot and negative at a 6 room knot. The renter room histogram declines smoothly through the would-be kink (4 rooms 27.0%, 5 rooms 17.3%, 6 rooms 10.1%, 7 rooms 4.4%), so there is no bunching at the cap.

Three readings follow. First, large rental units exist in large absolute numbers: 13.4 million occupied 3+ bedroom rentals nationally, which is not a number compatible with a literal prohibition. Second, they are overwhelmingly single-family houses rented out one at a time, not purpose-built family apartments; only 7.6% of units in 20+ unit buildings have three or more bedrooms. Third, the scarcity is a quantity phenomenon, not a price phenomenon: rent per bedroom and rent per square foot fall monotonically with size, and the within-market marginal price of family-sized space is statistically zero.

(d) What the literature says

Is the rental stock a different good?

The best direct evidence is Halket, Nesheim and Oswald (2020), which estimates a selection model of how English dwellings are allocated across the owner-occupied, private-rented and social sectors. Three of their results bear on our assumption. The owner-occupied share of the Greater London stock rises from 33.1% for dwellings under 50 square metres to 90.1% for dwellings over 100 square metres, with the private-rented share falling from 27.4% to 7.2%; the owner share is 94.4% for detached houses and 20.7% for high-rise flats; and, most usefully, unobserved quality in the rental sector declines with size, with an average quality gap of roughly 22% between a 50 and a 100 square metre rental property, a pattern absent on the owner side. That last finding is the strongest available support for treating large rental and large owner units as different goods rather than the same good under different tenure. It is estimated on English data, so transferring it to the United States is an assumption.

Glaeser and Shapiro (2003) supply the American counterpart and locate the segmentation on the structure-type axis rather than the size axis: 85.5% of people in single-family detached houses own, 85.9% of people in multi-unit buildings rent, and the explanation offered is the maintenance agency problem, since the major maintenance decisions in a multi-unit building are building- specific rather than apartment-specific and are therefore better made by a single owner. This is the empirical form of the Henderson and Ioannides (1983) rental externality: a tenant on a fixed-rent lease does not internalize the effect of use on the structure, so landlords bear a moral-hazard cost, and the cost is larger where the household's expected tenure is longer. Families are precisely the long-tenure households. I was not able to open Henderson and Ioannides directly, so that characterization is reported from secondary sources and should be checked before it appears in the manuscript.

Two candidate sources turn out not to speak to the question. Sinai and Souleles (2005) treat housing as a single good and argue for ownership as a hedge against rent risk; there is no discussion of stock heterogeneity. Bachmann and Cooper document gross flows between the renter and owner segments and say nothing about the physical composition of either stock.

How leading quantitative models handle the rental menu

Paper	Rental size object	How disciplined
Kaplan, Mitman and Violante (2020)	partial segmentation on a discrete menu: owner sizes {1.50, 1.92, 2.46, 3.15, 4.03, 5.15}, rental sizes {1.17, 1.50, 1.92}; the smallest class cannot be owned and the largest four cannot be rented	inside the calibration, targeting an owner/renter average size ratio of 1.5 and an owner/renter earnings ratio of 2.1
Greaney, Parkhomenko and Van Nieuwerburgh (2025), December 5 2025 draft	no size cap; owner sizes discrete on a six-point grid, rental size continuous, "maximum size is assumed to be the same as for owner-occupied houses: 12.7"	owner grid from ACS values and AHS size percentiles; no rental-bound row in the calibration table

Paper	Rental size object	How disciplined
Sommer and Sullivan (2018)	none; one linear technology converts a unit of housing stock into a unit of shelter services, so the two stocks are the same good and fully convertible, with household landlords	per-unit landlord management cost only
Landvoigt, Piazzesi and Schneider (2015)	no rental sector at all; continuous owner quality assigned in equilibrium	renting assumed away by appeal to the rarity of owner-to-renter transitions
Chambers, Garriga and Schlagenhauf (2009)	owner-side minimum size, the mirror object	estimated as part of the exactly identified problem
Halket and Pignatti (2015)	none; scarcity is an equilibrium search and screening outcome	Craigslist listings by zip and bedrooms, time on market, rent-to-price
Halket and Vasudev (2014)	none; the housing choice space is convex and does not depend on tenure. Renters end up in smaller units endogenously: average rental size 0.70 against average owned size 1.11	tenure sorting from down payments, transaction costs and tax wedges
Couillard (2025)	none; household type is the full cross of age, tenure, family status and size	rent coefficients estimated with additive heterogeneity by dimension

Kaplan, Mitman and Violante (2020) is the closest precedent and the one to cite, but it comes with two warnings. Their Table 6 reports that 10% of renters in the AHS occupy one of the top four size classes while the model puts 0% there by construction, which is the same failure our cap produces. And their Section 5.3.1 re-runs the model with no segmentation, half segmentation and full segmentation and reports results “virtually indistinguishable from baseline”, because in their setting renters are not constrained in housing services. Our model is the opposite case: the cap is a first-order lever. A referee who knows this literature will notice that we are leaning on a device its own originators showed to be inessential in their application, without having shown it is essential in ours for a reason other than assumption.

The reference urban model does not support the assumption. A July 2026 internal memo recorded that it sets the rental cap at 1.32 times the owner grid minimum, and the canonical $\bar{h}^R = 6$ was defended partly as that rule applied to our own sample (output/model/fable_size_mapping_audit_20260701/CAP_EXTERNAL_NOTE_20260706.md). The December 5, 2025 draft in the repository (/Users/tommasodesanto/Desktop/Projects/Fertility/Fertility_Spring26/DUE.txt, section 2.1.5 and the calibration table) says the opposite in plain text: “The lower bound of the size of a rental unit is zero and the maximum size is assumed to be the same as for owner-occupied houses: 12.7.” The calibration table lists the smallest and largest owner size and has no rental-bound row. The model’s rental and owner size sets do differ, but only in that one is discrete and the other continuous, which is a computational convenience the authors state as such. Either the earlier draft differed or the earlier memo misread it. Until that is resolved, the current draft cannot be cited as a precedent for tenure segmentation by size, and the external anchoring rule for $\bar{h}^R$ loses its stated pedigree.

Halket and Pignatti (2015) offer the microfoundation the hard cap is a reduced form of. Houses are ex ante identical; rental scarcity arises because landlords post lower rents but screen harder against long-duration households, so families face low matching probabilities rather than high posted rents. Thin quantities with flat prices is exactly the signature in our data. They describe the hard cap as the shortcut they replace, naming Chambers, Garriga and Schlagenhauf (2009), Fisher and Gervais (2011) and Amior and Halket (2014) as models that impose large houses being supplied only on the owner-occupied market.

Finally, the conversion evidence cuts against a literal cap. Kaplan, Mitman and Violante cite roughly 3 million single-family units converted from owner-occupied to rental use between 2007 and 2011 against 4.5 million owner-to-renter household transitions, and conclude that conversions account for the bulk of those transitions. Lambie-Hanson, Li and Slonkosky document the institutional side: institutional investors averaged 11.7% of single-family purchases in their sample and all investor purchases reached 38.1% of transactions, with the rise in institutional share explaining 58% of the increase in real house price growth in affected counties. Gurun, Wu, Xiao and Yang on the welfare effects of institutional landlords could not be opened and is marked not verified.

(e) Identification: what is established, what is conjectured

Established, in the sense that it is measured and reproducible:

1. The renter share falls steeply and monotonically in unit size, to roughly 9 to 11% at four or more bedrooms, in both AHS 2023 and ACS 2024, nationally and in New York and Los Angeles.
2. Within year by metro by location cells, the marginal rent of a family-sized room is statistically zero, and there is no bunching of renter unit sizes at any candidate cap.
3. Rent per bedroom and rent per square foot fall with unit size. Large rentals are cheap per unit of space, not expensive.
4. Family-sized rentals are concentrated in single-family structures; large multifamily buildings essentially do not contain them.
5. In England, where the question has been studied with a selection model, large rental units are systematically lower in unobserved quality than small ones, by about 22% between 50 and 100 square metres, while owner-occupied quality does not vary with size (Halket, Nesheim and Oswald 2020).

Conjectured, in the sense that the paper cannot currently demonstrate it:

1. That the observed size distribution of renters reflects a choice-set restriction rather than sorting. Renters are younger, poorer and more mobile than owners; all three predict small units without any constraint.
2. That $\bar{h}^R = 6$ is the right level. The current defense is that six rooms is the ninetieth percentile of renter unit sizes in the matched ACS sample. That is a description of the outcome, not identification: fitting a constraint to the upper tail of the distribution it is supposed to generate is circular, and the model then predicts zero renters above six rooms against 6.0% of prime-age childless renters observed at seven or more.

What would identify a constraint rather than a preference outcome, and what exists on each:

- **Rent per unit of size by size.** A binding quantity constraint with excess demand should show up as a premium on large rentals. It does not: the within-cell premium is zero and the gradient runs the other way. This is evidence *against* a price-rationed cap and *for* a non-price rationing story, but it is also consistent with no constraint at all.
- **Unobserved quality by size and tenure.** If large rentals are a genuinely worse good, a household wanting family space really does have to buy to get it, and the cap is a defensible reduced form. Halket,

Nesheim and Oswald

(2020) find exactly this, but on English data with an English social-housing sector. No American replication has been located. This is the single most valuable piece of evidence the paper could add.

- **Vacancy and time on market by size.** Under quantity rationing, large rentals should clear faster and sit vacant less. Halket and Pignatti measured time on market by zip and bedrooms with Craigslist listings and found screening against long-duration households. The equivalent tabulation for the current sample does not exist: I could not find a published Census Housing Vacancy Survey table by bedroom count, and the AHS Table Creator cross-tab has not been run here.
- **Conversion frictions.** The cleanest test is whether the stock moves between tenures at the family-sized margin. It does, at scale: roughly 3 million single-family units were converted from owner-occupied to rental use between 2007 and 2011, and institutional investors reached 11.7% of single-family purchases in the Lambie-Hanson, Li and Slonkosky sample. If large units were technologically owner-only, that conversion could not have happened. This is the strongest single argument against a literal cap and the best case for reading $\bar{h}^R$ as a friction that binds in the short run rather than a technological restriction.
- **Condominium conversion law and multifamily zoning.** These would give policy variation in the supply of large non-owner units. No evidence has been assembled here.

What the paper can claim: that the family-sized stock is, as a matter of measured fact, overwhelmingly owner-occupied; that this is a quantity fact and not a price fact; and that a hard cap is a standard reduced-form device for representing it, with Kaplan, Mitman and Violante (2020) as the closest precedent and Halket and Pignatti (2015) as the microfoundation. What the paper cannot claim: that the cap is identified, that it is a technological restriction, or that the reference urban model imposes one.

(f) Bottom line

The strongest defensible statement is that family-sized housing in the United States is overwhelmingly owner-occupied, that the scarcity of large rentals shows up in quantities and in unobserved quality rather than in posted prices, and that the model's hard cap at six rooms is a reduced form for a screening or availability friction of the Halket-Pignatti type, with the same status as the partial segmentation in Kaplan, Mitman and Violante (2020). The cap level itself is set to the ninetieth percentile of observed renter unit sizes, which is a calibration convention rather than an identified parameter, and the model's implication of zero renters above six rooms is false against a data share of 6.0% among prime-age childless renters. The weakest point, and the one a referee will attack first, is that the cap is the model's strongest fertility lever, since moving it from six to eight rooms raises the baseline total fertility rate by 0.071 and collapses the family-ownership gap from 0.296 to 0.035, so a first-order quantitative result rests on an unidentified assumption that the originators of the device themselves showed to be inessential in their own application. The second line of attack is conversion: roughly 3 million single-family units moved from owner-occupied to rental use between 2007 and 2011, which rules out any technological reading of the cap. The honest framing is to present the cap as a calibrated approximation to an availability and quality friction, report the fit over a band such as $\bar{h}^R \in \{5.5, 6.0, 6.5\}$, state the counterfactual implication the cap gets wrong, and let the mechanism rest on the down-payment threshold rather than on segmentation alone.

Not done

- Henderson and Ioannides (1983) could not be opened; JSTOR blocked access and no working-paper mirror was found. The rental-externality characterization above is from secondary sources, principally Glaeser and Shapiro (2003).

- Gurun, Wu, Xiao and Yang on institutional landlords and renter welfare could not be opened past the paywall. Its reported magnitudes are not verified.
- Glaeser and Gyourko's *Rethinking Federal Housing Policy* was not opened; Glaeser and Shapiro (2003) was opened instead and carries the stylized fact.
- No vacancy rate or time-on-market tabulation by bedroom count was obtained. Census Housing Vacancy Survey historical tables break out vacancy by units-in-structure and region, not bedrooms; AHS Table Creator was not run.
- The ACS 2024 figures were pulled through the Census Reporter API rather than data.census.gov or the Census API directly, and are 1-year rather than 5-year estimates. They should be re-pulled from the primary source before print.
- The internal discrepancy on the reference urban model's rental cap is reported, not resolved. Resolving it requires locating the earlier draft that the July 2026 memo read.
- No American replication of the Halket-Nesheim-Oswald quality-by-size-and- tenure result was located. This is the highest-value missing evidence.

References

Each entry states whether the source was opened at first hand for this note, opened by a delegated agent that reported the extracted text, or verified earlier in the repository's own July 2026 memo.

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